

SAMUEL FRAUSTO

Creative Technology & Interactive Design

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— EDUCATION

B.S. Human Technology Interaction

USC Iovine and Young Academy

Jan 2025 - May 2028

GPA: 3.67 | Expected graduation: May 2028

- Inaugural graduating cohort; interdisciplinary degree offered by Iovine and Young Academy in partnership with the Viterbi School of Engineering.
- Coursework: Extended Reality Development, Collaborative Prototyping for Healthcare Innovation, Constructing Digital Worlds; completed CSCI 103 in C++.

— PROFESSIONAL EXPERIENCE

Extended Reality Learning Design Intern

Edwards Lifesciences

Jun 2026 - Present

Continued as a part-time contractor

- Lead experience design and prototype implementation for immersive training that helps engineers practice clinical conversations before handling patient calls.
- Translate clinical workflows and communication goals into interactive scenarios, feedback logic, and repeatable training experiences.
- Designed and built Heart, an interactive web experience explaining cardiac anatomy, blood flow, and valve mechanics to non-specialist learners.
- Selected to continue after the internship on a 150-hour part-time contract through December 2026, carrying training projects from design through implementation.

Media & Communications Intern

The Jel Sert Company

Jun 2025 - Dec 2025

- Produced launch media, communications, and campaign creative for Crayola x Fla-Vor-Ice, Pure Kick x Harley-Davidson, and Jamba Singles To Go.
- Managed 30+ partnership and influencer contacts for the Jamba launch and represented the company at a California road show.
- Wrote press and launch content, produced promotional creative, and built internal campaign-performance reporting to support creative decisions.

Research & Development Lab Intern

The Jel Sert Company

Jun 2024 - Aug 2024

- Maintained stability studies for 100+ products through pH, penetration, color, and sensory testing in shelf and controlled environments.
- Tested artificial-sweetener formula combinations across four products and identified optimal formulations through structured experiments.

ResearchHStart Fellowship Awardee

University of Chicago | Kron Lab

May 2023 - Jul 2023

- Completed 300+ hours of independent wet-lab research evaluating small-molecule TET2 inhibition in CT-26 cell cultures as a potential response to adaptive radiation resistance.
- Analyzed experimental data and presented the study to faculty at the UIUC cancer research symposium, placing 2nd among 36 presenters.

SKILLS

Design & prototyping

Interaction design, systems design, learning design, interactive prototyping

3D & XR

Unity, Blender, NVIDIA Omniverse, USD, Meta Quest, Apple Vision Pro

Implementation

C# scripting, JavaScript, HTML/CSS, HLSL, VS Code

Creative tools

Figma, Photoshop, DaVinci Resolve, Canva

Research

Experiment design, data analysis, research communication

Foundational exposure

C++, Python, Swift

Additional engine

Unreal Engine

AI-assisted coding tools

Claude, Codex, Sakana AI, Cursor

CERTIFICATION

Lean Six Sigma Yellow Belt

Earned through an immersive clinical-training project at Edwards Lifesciences | Aug 2026

PORTFOLIO



samfrausto.com

Projects & case studies

PROJECTS

Autonomous Traffic Digital Twin

USC Iovine and Young Academy Aug 2026 - Present
USC faculty-directed research | In progress

- Developing an interactive traffic sandbox in NVIDIA Omniverse to explore coordinated autonomous vehicles and traffic interventions.
- Approved scope includes an OpenUSD digital twin, virtual instrumentation, and live parameter controls; research and implementation are in progress.

Iovine & Young Hall Digital Twin

USC Iovine and Young Academy Jan 2026 - Present

- Developed the delivery layer and waypoint navigation on a three-person web team for a classwide digital twin of Iovine and Young Hall.
- Helped deliver a shared Omniverse model across web, Meta Quest, and Apple Vision Pro with continuous transitions between viewpoints.

VR Baseball Training System

Independent Feb 2025 - Present
Independent project | In progress

- Developed a VR baseball training prototype in Unity/C#, including the pitcher's throwing mechanic, character rigging in Unity, and different pitch types.
- Created 3D models in Blender and assembled the experience for Meta Quest, focusing on ball movement, realism, and usability.

SuzChews

USC ACAD 432 Jan 2026 - May 2026
Research-informed product concept

- Translated clinical research on chemotherapy-related taste distortion into a gum product concept with four active-ingredient strategies.
- Designed flavor-release logic, product architecture, branding, and an interactive explanation of the proposed patient experience.

Cyberpunk 2077 Room Digital Twin

USC ACAD 288 Aug 2025 - Dec 2025

- Co-led 3D modeling, animation, and skybox production for a room digital twin in NVIDIA Omniverse.
- Built an Omniverse extension to remove a repetitive manual step from the team's Blender-to-USD material-transfer workflow.

Synesthesia: Ambient visionOS App

USC ACAD 217 Feb 2025 - May 2025

- Led interaction and runtime development for an ambient visionOS app that changes scenes behind a user's active workspace.
- Built scene, color, music, and AR-opacity controls; brought the Unity prototype to Vision Pro with custom gradient shaders.

Pavilia: Portal Mechanic

USC ACAD 217 Feb 2025 - May 2025

- Built a perspective-correct Unity portal using dual-camera transforms and live render textures between two student-built worlds.
- Reworked the HLSL projection shader to prevent edge clipping and implemented seamless player transfer through the portal.

ADDITIONAL RESEARCH

AI vs. Expert Proteomics Analysis

Independent | UChicago
Jul 2023 - Aug 2023

- Collaborated with the UChicago Proteomics Core lead to compare AI-assisted and traditional proteomic data-sorting approaches.
- Analyzed 2,100+ protein sequences across 80+ pathways and presented findings to the principal investigator and research team.

LEADERSHIP

Programming Chair & Team Captain

USC Club Baseball
Jan 2025 - Present

- Lead teammates while competing against colleges nationwide.
- Plan philanthropy and social events and produce first-person game videos using an on-field GoPro.

STUDY ABROAD

Franklin University Switzerland

Non-degree study | Lugano
Aug - Dec 2024
20 units | GPA: 4.0

RECOGNITION

Fleur de Lis Award

Selected as 1 of 12 recipients from a class of approximately 750 for academics, character, and leadership | St. Charles East High School, 2024

LANGUAGES

English / Spanish
Native / Basic